

- 8 (a) State what is meant by the *binding energy* of a nucleus and how it is related to the mass defect.

Binding energy is the energy required to separate to infinity all the constituent nucleons in the nucleus.

It is (the energy equivalent of mass defect and) related to mass defect in the equation Binding energy = (mass defect) $\times c^2$

[2]

- (b) The binding energy graph on Fig. 8.1 shows the variation with nucleon number A of the binding energy per nucleon. Some common nuclides are plotted on the graph, with a few of them labelled as shown.

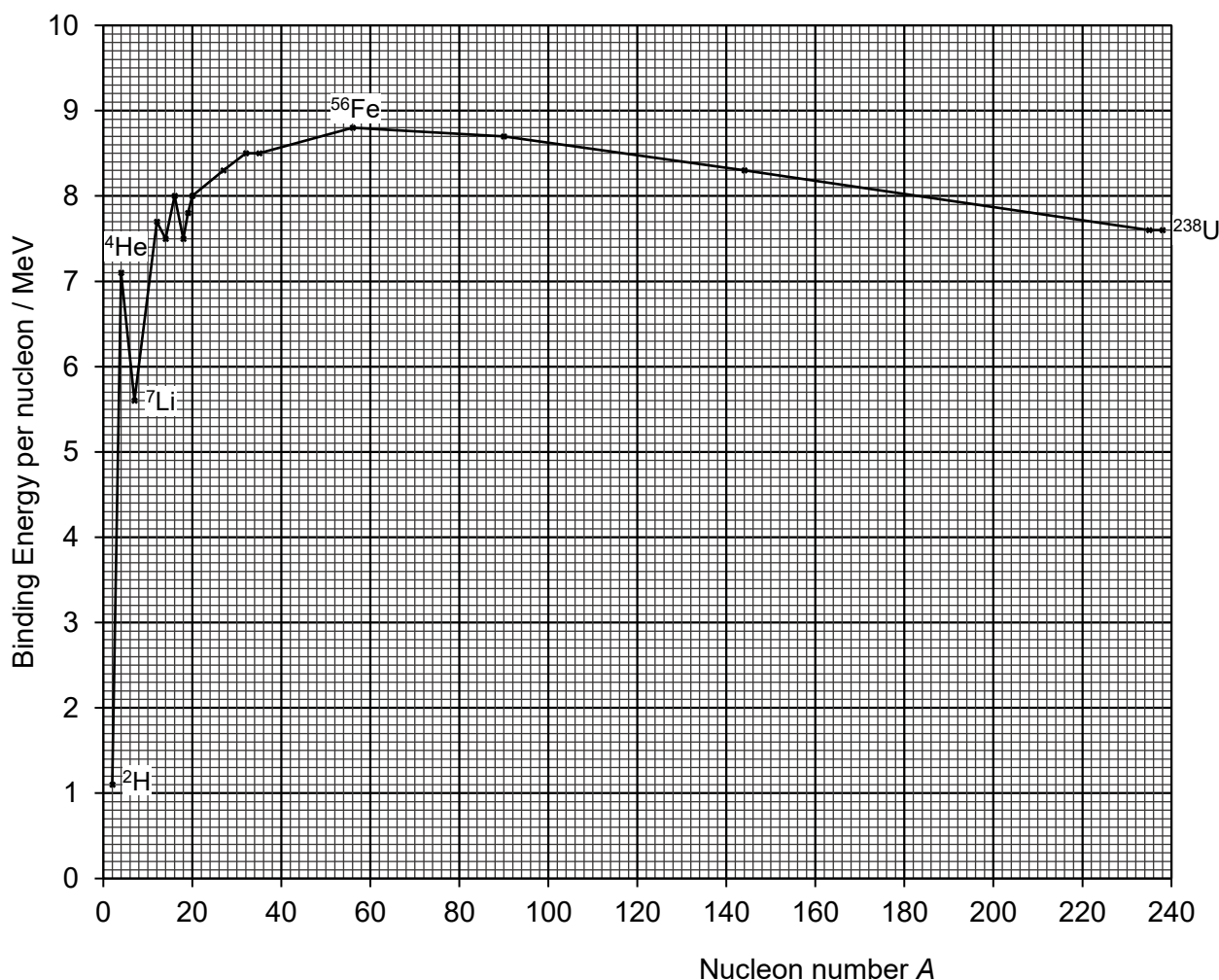


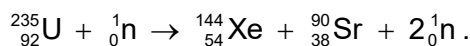
Fig. 8.1

- (i) Explain why hydrogen-1 is not typically included in a binding energy graph.

H-1 is a proton which is a nucleon/constituent particle of the nucleus. Hence it has no binding energy or there is no meaning to binding energy of a proton. (identify & reasoning) [1]

[1]

- (ii) A nuclear power station uses uranium-235 as fuel in fission reactions. One possible fission reaction is



1. Use data from Fig. 8.1 to show that energy released in the reaction is about 190 MeV.

$$\begin{aligned} \text{Energy released} &= \text{total final BE} - \text{total initial BE} \\ &= (144 \times 8.3) + (90 \times 8.7) - (235 \times 7.6) \quad [1] \\ &= 192.2 \text{ MeV} \quad [1] \\ &= 190 \text{ MeV} \end{aligned}$$

[2]

2. Hence, calculate the energy released in the fission of 1 kg of uranium-235.

$$\begin{aligned} \text{Energy released} &= \text{no of reactions} \times 190 \text{ MeV} \\ &= (1/235\text{u}) \times 190 \text{ MeV} \quad [1] \\ &= 4.9 \times 10^{26} \text{ MeV} \quad [1] \end{aligned}$$

energy released = MeV [2]

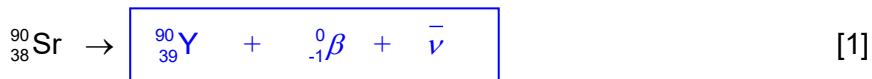
- (c) A small sample of waste produced by the reactor in (b)(ii) contains strontium-90 (${}_{38}^{90}\text{Sr}$). Strontium-90 is radioactive and undergoes beta decay into a daughter nuclide Yttrium-90 (Y).

- (i) In beta decay, it was discovered that an antineutrino ($\bar{\nu}$) must be emitted in order that two conservation laws are not violated. State the two conservation laws.

Conservation of momentum and Conservation of energy/mass-energy.

.....[1]

- (ii) Complete the beta decay equation, including all the decay products.



(Optional to include antineutrino)

- (iii) A radiation detector is placed close to the sample to measure the count rate for strontium-90 found in the sample. Fig. 8.2 below shows the variation with time t of the count rate.

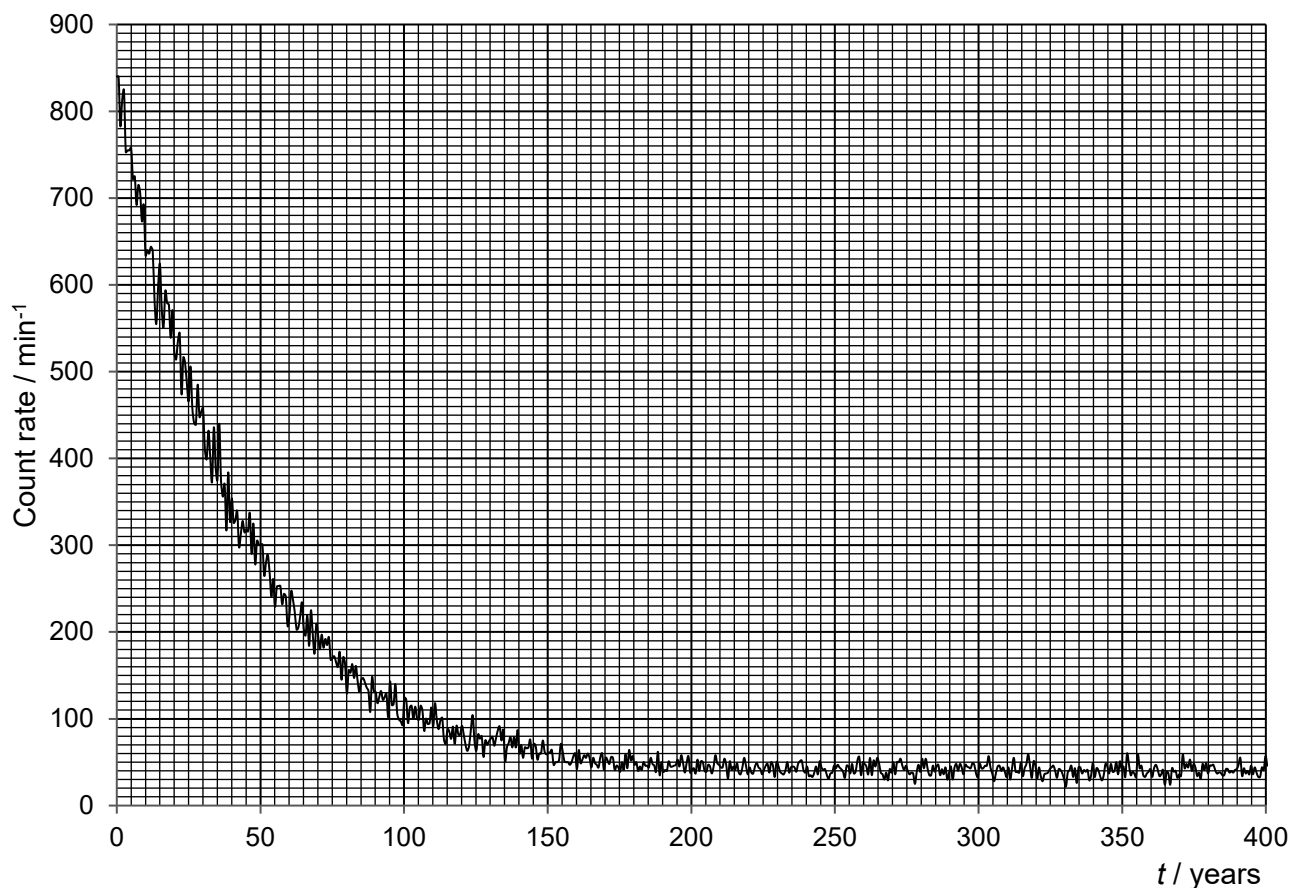


Fig. 8.2

1. State the feature of Fig. 8.2 that indicates the random nature of radioactive decay.

Fluctuations on the curve / curve is not smooth

.....[1]

2. Use Fig. 8.2 to determine the half-life of strontium-90.

Background count rate = 40 min^{-1} considered backgrd in calculation of $t_{1/2}$ (allow 30-50) [1]

$t_{1/2, 1}$

Actual count rate at $t=0 = 840 - 40 = 800 \text{ min}^{-1}$

After 1 $t_{1/2}$, actual count rate = 400 min^{-1}

measured count rate = $400 + 40 = 440 \text{ min}^{-1}$

From graph, $t_{1/2} = 35 - 0 = 35 \text{ yrs}$

method to determine $t_{1/2}$ [1]

$t_{1/2, 2}$

Actual count rate at $t=50 \text{ yr} = 300 - 40 = 260 \text{ min}^{-1}$

After 1 $t_{1/2}$, actual count rate = 130 min^{-1}

measured count rate = $130 + 40 = 170 \text{ min}^{-1}$

From graph, $t_{1/2} = 75 - 50 = 25 \text{ yrs}$

Average $t_{1/2} = (35 + 25)/2 = 30 \text{ years}$

determined half-life at least twice and find $\langle t_{1/2} \rangle$, range within 26 – 32 yrs [1]

half-life = years [4]

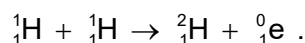
3. Hence, determine the probability that a nuclide of strontium-90 will undergo decay in 1 year.

$$\begin{aligned}
 \text{Probability of decay in 1 yr} &= \text{Probability of decay per unit time} \times 1 \text{ yr} \\
 &= \lambda \times 1 \text{ yr} \\
 &= (\ln 2 / t_{1/2}) \times 1 \text{ yr} \\
 &= (\ln 2) / (30 \text{ yr}) \times 1 \text{ yr} \\
 &= 0.023 \text{ or } 2.3\%
 \end{aligned}$$

[1]

probability =[1]

- (d) Nuclear fusion occurs in the core of stars composed of ionised gas. A possible fusion reaction is



Each ${}_1^1\text{H}$ nuclide can be considered to be a sphere of radius 0.0010 pm. Fusion occurs when the two nuclides are able to overcome the force of repulsion between them and collide.

- (i) Show that the minimum total kinetic energy required of the two ${}_1^1\text{H}$ nuclides for fusion to occur is $1.2 \times 10^{-13} \text{ J}$.

By cons. of energy,

Total initial KE = Total final EPE

$$\begin{aligned}
 &= (+e)(+e) / (4\pi\epsilon_0 d) \\
 &= (1.6 \times 10^{-19})^2 / (4\pi\epsilon_0 \times 2 \times 0.0010 \times 10^{-12}) \\
 &= 1.15 \times 10^{-13} \text{ J} \\
 &= 1.2 \times 10^{-13} \text{ J}
 \end{aligned}$$

[1]

[1]

[2]

- (ii) If the ionised gas is assumed to be ideal, determine the temperature of the gas required for fusion to occur.

Average KE of nuclide = $3/2 kT$

$$(1.2 \times 10^{-13}) / 2 = 3/2 kT \quad [1]$$

$$T = 2.9 \times 10^9 \text{ K} \quad [1] \quad (\text{if used intermediate K, } T = 2.8 \times 10^9 \text{ K})$$

temperature = K [2]

- (iii) The temperature of the core of the Sun is known to be about $1.5 \times 10^7 \text{ K}$. With reference to (d)(i) and (d)(ii), comment on the actual kinetic energy of the nuclei in the Sun's core.

Actual temperature of Sun is much cooler than temp calculated in (ii)

So Sun's core must involve nuclei with actual KE far above the average KE calculated in (i) (since fusion must occur). [1]

[Total: 20]

- 9 (a) State what is meant by a *field of force*